ETHAN DEBEB

Atlanta Metro Area | LinkedIn | Digital Portfolio

EDUCATION

Intended Major: **Aerospace Engineering,** Purdue University, West Lafayette, IN Expected: May 2030

Wheeler High School STEM Magnet | Marietta, GA | Expected: May 2026 | GPA 4.0 | ACT Score: 36
Relevant Coursework: Adv. Physics Robotics, AP Physics C: Mechanics, AP Physics I & II, AP Computer Sci Principles

TECHNICAL SKILLS

Programming: Python, Java, C++
Software: SolidWorks, OnShape CAD, Google Suite, MS Office Suite, VS Code
Languages: Amharic and Spanish

EXPERIENCE

Kennesaw State University AERO Lab, Marietta, GA July 2025 - Present
Aerospace Research Intern

- Researching and innovating Unmanned Aircraft Systems (UAS) and Multidisciplinary Design Optimization.
- Researched integrating and assessing YOLO object detection on UAVs to count vehicles
- Created a Life Jacket Delivery Mechanism for UAV

Publication*:*
Adeel Khalid, Owais Ahmed and Ethan Debeb. "Real-Time Vehicle Detection and Enumeration Using Unmanned Aerial Systems," AIAA 2026-4753. *AIAA AVIATION 2026 Forum*. June 2026. DOI: https://doi.org/10.2514/6.2026-4753

LEADERSHIP

Students In Space Club
Co-President SY 2025-2026 **Vice-President** SY 2024-2025

- Serve as a bridge between members and the leadership team by assisting the club's president with planning and initiatives.
- Organize meetings, coordinate aerospace projects, and prepare for competitions.
- Assist with project design and mentoring members, build and present for competitions. Manage the Students in Space Podcast.

Competitions: 2025 Canada Space Design Competition, 2024 Georgia Tech/NASA Payload Challenge

CircuitRunners Robotics, Inc
FIRST Tech Challenge - 1002 Surge SY 2022-Present

- **Marketing Lead:** Manage and provide strategic marketing direction in collaboration with FTC 1002 team members to promote robotics to external audiences.
- **Designer & CAD Builder Sub-Team Member:** Assist with planning, designing, & the mechanisms for the robot. Led team to two qualifying world championships and three state championships. Mentor freshmen and others in optimizing skills in building accuracy, troubleshooting issues, and integrating builds with software.
- **Drive Coach Sub-Team Member:** Develop competition strategies to gain more points that comply with game rules. Conduct or attend strategy meetings to analyze how other teams are employing their robots. Instruct the robot driver on how to strategize with other teams during matches.

FIRST Robotics World Championship Appearances*: 2026, 2024, 2023*
Awards*: Georgia Competition: 2025 2nd Inspire, 2023: 3rd Inspire, 2022: Inspire & Winning Alliance, Alabama State Championship: 2022: Innovate Award, 2023 FIRST Championship: Innovate Award sponsored by Raytheon Technologies*

MEMBERSHIP

National Society of Black Engineers (NSBE)
American Institute of Aeronautics and Astronautics (AIAA)